Exploring predictors of crime rates across Victorian LGAS from 2011-2024

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Acknowledgement

This project explores predictors of crime rates across Victorian Local Government Areas (LGAs) from 2011 to 2024. It was undertaken as a student research project through the Monash Student Analytics Club (MSAC), with the aim of applying quantitative and econometric methods to a real-world policy and social issue. The authors are student researchers, and this paper represents an independent academic exercise rather than an official publication or position of MSAC or Monash University.

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The views and interpretations expressed in this project are solely those of the authors and should not be attributed to MSAC, Monash University, or any other affiliated organisation. Any remaining errors are the authors' own.

Abstract

This paper examines the determinants of crime across Victorian Local Government Areas (LGAs) using panel data from 2011 to 2024. The analysis incorporates a range of demographic, economic, and geographic variables, including education attainment, income, unemployment, fertility, population structure, and citizenship composition. An array of econometric models is applied, including pooled OLS, fixed effects models, factor analysis, and regularisation techniques such as LASSO and Elastic Net, to address unobserved heterogeneity and multicollinearity.

The results show that crime is influenced by a combination of socio-economic and demographic factors rather than a single dominant variable. Across models, fertility rate and education are consistently associated with lower crime**,** while unemployment and population density show positive relationships**.** Fixed effects models highlight the importance of within-LGA dynamics, demonstrating that some relationships observed in simpler models are driven by cross-sectional differences rather than changes over time. Regularisation methods confirm the relative importance of key predictors and improve model stability.

Among the models, Elastic Net provides the strongest predictive performance, effectively identifying the most relevant drivers while reducing the influence of weaker variables. Overall, the findings suggest that demographic structure and labour market conditions play a central role in explaining variation in crime across LGAs, with important implications for targeted policy design.

1. Introduction

Understanding the drivers of crime is important for both policy and resource allocation, particularly in a diverse and rapidly changing region such as Victoria. Crime rates vary significantly across Local Government Areas (LGAs), reflecting differences in demographic composition, economic conditions, and social structure. Identifying which of these factors are most strongly associated with crime is therefore essential for developing targeted and effective responses.

This paper examines the relationship between crime rates and a range of socio-economic and demographic variables across Victorian LGAs over time. The analysis focuses on key factors including age structure, education attainment, income, unemployment, fertility, population density, citizenship composition, and geographic proximity to the Central Business District (CBD). These variables are chosen to capture both economic conditions and underlying population characteristics that may influence crime outcomes.

A range of econometric models are applied to explore these relationships from different perspectives. Pooled Ordinary Least Squares (OLS) models provide a baseline comparison, while fixed effects models control for unobserved differences across LGAs and over time. Factor analysis is used to examine the structure of the data and identify potential multicollinearity between variables. To address this issue and improve variable selection, regularisation techniques including LASSO and Elastic Net are employed.

By combining these approaches, the paper aims to identify the most consistent and robust predictors of crime. The results show that crime is influenced by a combination of socio-economic factors rather than a single dominant variable, with education, fertility, and labour market conditions emerging as key contributors across multiple models.

2. Literature Review

A large body of literature highlights that crime is influenced by a combination of socio-economic, demographic, and environmental factors. Prior research provides a strong basis for the selection of variables used in this paper and supports the relevance of analysing crime at the LGA level.

Economic and labour market conditions are consistently identified as key drivers of crime. Studies show that unemployment, income, and broader economic conditions influence both short-run and long-run crime dynamics, with higher unemployment generally associated with increased crime rates (Lau, 2013).However, the relationship is complex, with some evidence suggesting that economic adversity interacts with social conditions rather than acting as a standalone driver (Economic Transformation and Regional Crime, 2001) These findings justify the inclusion of unemployment rate and median income as key explanatory variables.

Education is also widely recognised as an important determinant of crime. Higher levels of educational attainment are generally associated with lower crime, as education improves employment opportunities and reduces incentives for criminal activity (Joel Carr, 2022).This supports the use of education variables, particularly bachelor's degree attainment, as indicators of socio-economic status.

Demographic structure plays a significant role in shaping crime patterns. Age composition and gender differences influence the likelihood of crime (Clare & Morgan, 2009). In addition, fertility and broader social structure have been linked to crime outcomes, with evidence suggesting that areas with higher fertility rates may reflect more stable, family-oriented environments and lower crime levels (Lau, 2013) These findings justify the inclusion of weighted average age and fertility rate.

Population size, density, and geographic accessibility are also important determinants of crime. Research on regional Australia shows that population concentration and accessibility to services influence local crime levels by shaping economic activity and social interactions (Carcach, 2000). Similarly, proximity to urban centres, such as distance to the CBD, captures spatial differences in opportunity and exposure to crime.

Citizenship and ethnicity-related variables have also been explored in the literature. Studies using LGA-level data in Australia find that citizenship composition is associated with variations in crime rates, although these relationships are complex and require careful interpretation (Francis, Armstrong, & Totikidis, 2006). This supports the inclusion of the percentage of non-Australian citizens as a demographic indicator.

Environmental and contextual factors further influence crime patterns. For example, temperature has been shown to affect crime rates, with higher temperatures associated with increases in certain types of crime (Heather R. Stevens, 2019).While such variables are not directly included in this paper due to data limitations, they highlight the broader range of influences on crime.

Overall, the literature suggests that crime is driven by a combination of economic conditions, demographic structure, and geographic factors, rather than a single variable. This reinforces the relevance of the current study, which examines multiple dimensions simultaneously using LGA-level panel data. By applying a range of econometric models, this paper contributes to the literature by identifying the most consistent and robust drivers of crime within the Victorian context.

3. Context and Data

This study uses a panel dataset of Victorian LGAs covering the period 2011 to 2024. The dataset combines multiple sources to capture demographic, economic, and geographic characteristics relevant to crime. Key variables include population size, population density, age structure, education attainment, income, unemployment, fertility, citizenship composition, and proximity to the CBD. Crime data is measured at the LGA level, allowing for both cross-sectional and time-series analysis.

The dataset is constructed from a combination of Australian Bureau of Statistics (ABS) census data, administrative datasets, and publicly available economic indicators. A detailed breakdown of data availability and missingness across variables is provided in Table 01 (found in the Appendix), which summarises the timing, gaps, and treatment of each variable.

Census-based variables, including education attainment and citizenship composition, are only available for the years 2011, 2016, and 2021. To construct a balanced panel, these values were extended to non-census years using the last observed value within each LGA. This approach assumes that structural demographic characteristics evolve gradually over time and is standard in panel data settings where high-frequency data is unavailable. To validate this assumption, pooled OLS models using the full dataset were compared with census-only regressions.

Other variables are available at different frequencies, with several containing gaps in earlier years. For example, income data is unavailable prior to 2016, fertility measures are only available from later periods, and gender composition data is only available from 2019 onwards. As outlined in Table 01, these inconsistencies were addressed by aligning variables to a common yearly frequency and retaining observations where sufficient data was available. Variables with substantial missingness or limited time coverage were excluded from the main analysis to maintain consistency across LGAs and time.

Several transformations were applied to improve comparability across LGAs. Where possible, raw counts were converted into rates or percentages to account for differences in population size. In models where raw offence counts are used, corresponding predictors are also expressed in raw terms to maintain scale consistency. Time-invariant variables, such as distance to the CBD and

geographic area, were excluded from fixed effects models as they are fully absorbed by LGA-specific effects.

To address multicollinearity, overlapping variables were removed or consolidated. Detailed education categories were reduced to representative levels, and citizenship counts were replaced with percentage measures where appropriate. Similarly, highly related demographic variables were excluded where they did not provide additional information beyond existing measures.

Finally, unemployment data required additional preprocessing, as it is reported at a regional level rather than by LGA. LGAs were mapped to their corresponding regions to assign unemployment rates consistently across the dataset.

Overall, the data cleaning process ensures a consistent and comparable panel dataset. While some interpolation and exclusion decisions are necessary, the resulting dataset is suitable for analysing the relationship between crime and socio-economic factors across Victorian LGAs.

4. Exploratory Data Analysis

This section presents an exploratory analysis of the spatial, distributional, and temporal patterns in the data. The objective is to characterise the structure of the cross-sectional and panel variation prior to model specification, and to flag distributional features and visual associations that motivate the modelling choices made in Section 5.

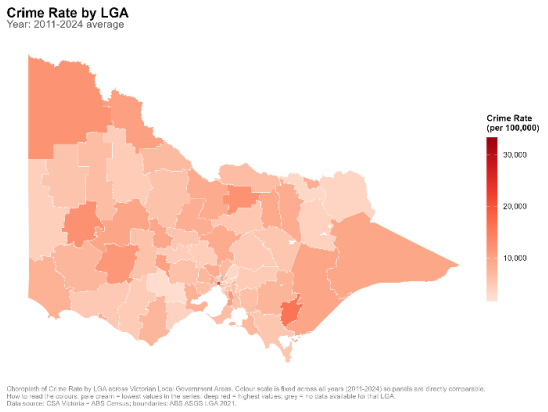


Figure 02

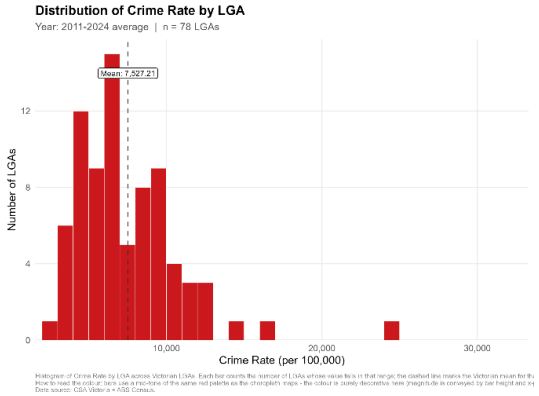


Figure 03

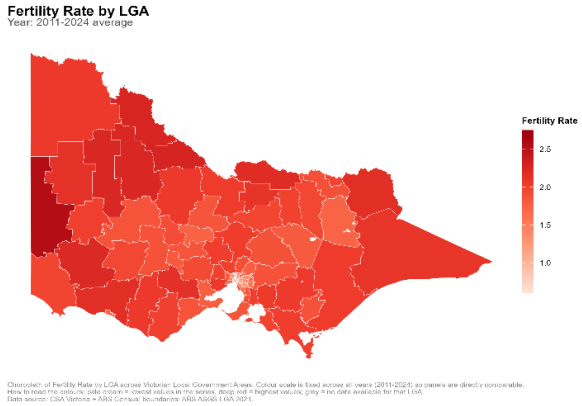


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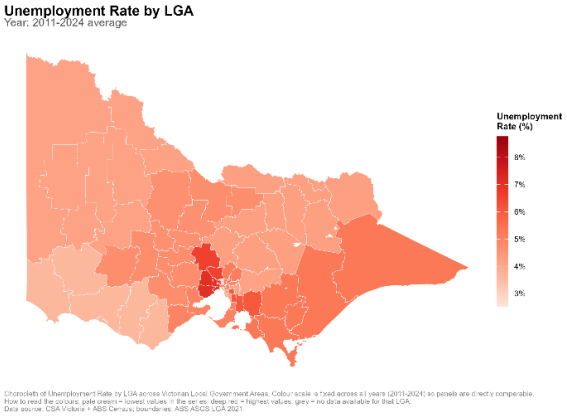


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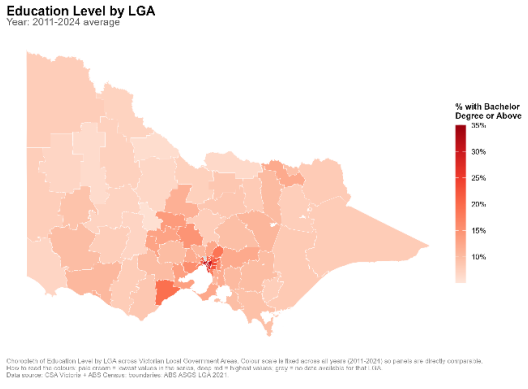


Figure 06

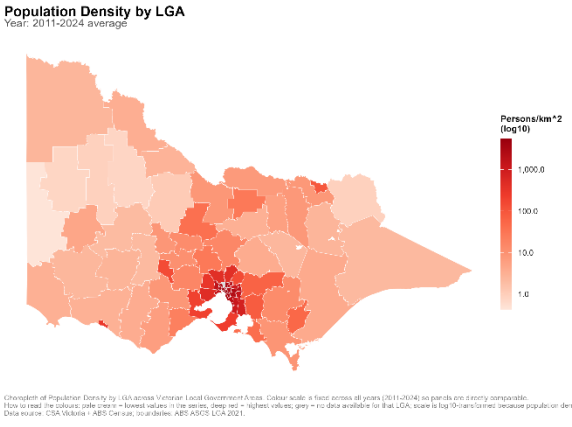


Figure 07

4.1 Spatial Distribution of Crime Rates

The choropleth map of recorded offences per 100,000 population, averaged across 2011 to 2024 (Figure 02)**,** reveals that crime in Victoria is far from uniformly distributed. The darkest shading - corresponding to rates well above the state average - is concentrated in a small cluster of inner Melbourne LGAs, most prominently the central business district and its immediate surroundings, 11 where average rates exceed roughly 15,000 offences per 100,000. A second, geographically isolated pocket of elevated crime is observed in the far north-west of the state, where a single large rural LGA records rates comparable to the inner-metropolitan peak. Outside these hotspots, much of regional and rural Victoria, particularly the central west and the eastern alpine belt, displays markedly lower rates, with most LGAs falling in the range of roughly 3,000 to 8,000 offences per 100,000, while outer-suburban LGAs occupy an intermediate band. Taken together, the pattern appears to be spatially clustered rather than dispersed, with high-crime and low-crime LGAs each forming visually contiguous neighbourhoods. This feature may indicate the presence of spatial autocorrelation in the residuals of any cross-sectional specification and warrants further investigation, although a formal spatial econometric treatment lies beyond the scope of the present study.

4.2 Distribution of Crime Rates Across LGAs

The histogram of LGA-level crime rates averaged across 2011 to 2024 (Figure 03) indicates that the cross-sectional distribution is right-skewed rather than approximately normal. The mass of the distribution lies between roughly 2,000 and 12,000 offences per 100,000, with a modal bin near 6,000 to 7,000, while a long upper tail extends past 20,000. The mean crime rate is 7,527.21 offences per 100,000, which sits visibly to the right of the modal bin, a configuration consistent with positive skew. A total of 31 LGAs fall above the mean and 47 fall below it, with the median lying below the mean as expected under right-skew. Three clearly separated bars at the upper end of the distribution, situated near 14,000, 16,000, and 24,000 offences per 100,000, identify three LGAs as visible outliers. Cross-referencing these extreme values against the spatial map (Figure 02) suggests that they correspond predominantly to inner-metropolitan LGAs, although the most extreme outlier reflects a sparsely populated rural LGA in the north-west, where small

population denominators can amplify rate volatility. The skew implies that a logarithmic transformation of the crime rate may be appropriate prior to parametric modelling - a consideration that motivates the log-crime specification adopted in the Elastic Net model in Section 5.7.

4.3 Socioeconomic Indicators by LGA

4.3.1 Fertility Rate

The fertility rate map (Figure 04), reporting the LGA average over 2011 to 2024, displays values ranging from approximately 1.0 to above 2.5 births per woman. The lowest fertility rates appear to cluster tightly within inner Melbourne, where values fall below 1.5, while the highest rates are observed in regional and rural LGAs, particularly in the north-west and the eastern Gippsland region. Visually, this pattern is broadly the inverse of the crime map, in that the inner metropolitan areas with the highest crime rates also exhibit the lowest fertility rates. The relationship is not, however, monotonic across the state: the elevated-crime LGA in the northwest also records relatively high fertility, indicating that the bivariate association between crime and fertility may be non-linear and sensitive to the inclusion of metropolitan controls. These 14 patterns is consistent with the negative coefficient on fertility rate observed across the pooled OLS, LASSO, and Elastic Net specifications reported in Section 5.

4.3.2 Education Level

The education map (Figure 05), again averaged across 2011 to 2024, displays a pronounced metropolitan-regional gradient. The share of residents holding a Bachelor's degree or above peaks at over 35 per cent in inner-Melbourne LGAs, which appear in bright yellow, and declines steeply outside the metropolitan fringe, where most regional and rural LGAs fall below 15 per cent. The pattern is highly clustered geographically. When juxtaposed with the crime map (Figure 02), the inner-Melbourne hotspot is simultaneously high-crime and high-education, while regional Victoria is generally low on both. This co-occurrence appears to suggest that the 15 bivariate relationship between education and crime at the LGA level is confounded by metropolitan status, and may therefore not exhibit the negative direction commonly assumed at the individual level until cross-sectional differences between LGAs are controlled for. This

visual intuition aligns with the sign reversal of the bachelor's degree coefficient between the Time Fixed Effects and Combined Fixed Effects specifications reported in Section 5.

4.3.3 Unemployment Rate

The unemployment map (Figure 06), averaged across 2011 to 2024, shows rates ranging from below 3 per cent to above 8 per cent. The highest unemployment rates are concentrated in a band of inner and middle-ring metropolitan Melbourne LGAs, with several outer south-eastern LGAs 16 also displaying elevated values. Western and central regional Victoria, by contrast, records the lowest rates. The spatial alignment with the crime map is comparatively close: many of the LGAs identified as crime hotspots in Figure 02 also fall within the darker bands of the unemployment map, suggesting that unemployment may share a more direct visual association with crime than fertility or education. It should be emphasised, however, that the unemployment series is reported at the regional level and mapped to LGAs (as discussed in Section 5), so the within-region homogeneity visible in Figure 06 partly reflects the level of aggregation rather than true LGA-level variation.

4.3.4 Population Density

The population density map (Figure 07), averaged across 2011 to 2024 and plotted on a log10 scale, ranges from below 1 person per square kilometre in the alpine and Mallee regions to above 1,000 persons per square kilometre in inner Melbourne. The metropolitan-regional contrast is the sharpest of any indicator examined and aligns closely with the crime map: the densest LGAs are also among the highest-crime, while sparsely populated rural LGAs generally record below average crime rates. Population density therefore appears to be a strong visual correlate of crime at the LGA scale, although the strength of its co-movement with education and unemployment raises the prospect of multicollinearity in any joint specification, motivating the use of regularisation methods in Section 5.

4.4 Temporal Variation in Indicators

Although the figures presented in this section deliberately abstract from year-on-year movements by reporting LGA averages over 2011 to 2024, a brief comment on the underlying temporal structure is warranted. Across the panel, the Victorian average crime rate oscillates between approximately 6,500 and 8,500 offences per 100,000, with the cross-LGA distribution retaining its right-skewed form throughout the period and the spread across LGAs widening only modestly over time. Most LGAs trace broadly parallel trajectories around the state mean, indicating that LGA rankings are largely persistent and that reshuffling between high- and low-crime quantiles is limited. The persistence of LGA-level differences over time provides preliminary support for the use of LGA fixed effects in Section 5, while the comparatively muted year-on-year movements suggest that time fixed effects are likely to absorb common shocks without removing substantive variation. Year-by-year choropleths and histograms for each indicator are retained in the project's figure repository for completeness but are not reproduced here, as the focus of this exploratory analysis is on the long-run cross-sectional structure summarised in the averaged figures.

5. Methodology

This paper applies a sequence of econometric models to examine the relationship between crime and socio-economic factors across LGAs, with each model addressing a different limitation of the data.

The analysis begins with pooled OLS to establish baseline relationships between crime rates and key demographic and economic variables. This is followed by fixed effects models, which control for unobserved heterogeneity across LGAs and over time, allowing the analysis to focus on within-LGA changes rather than cross-sectional differences.

To further refine the analysis, time fixed effects are introduced to remove common shocks affecting all LGAs, while the combined fixed effects model controls for both LGA-specific and time-specific factors, providing the strongest identification of underlying relationships.

Factor analysis is then used to examine the correlation structure of the data and identify potential multicollinearity among variables. Given the presence of highly correlated predictors, regularisation methods are applied. LASSO is used to perform variable selection and identify the most important predictors, while Elastic Net extends this by retaining groups of correlated variables and improving model stability.

Together, this modelling approach allows for a comprehensive analysis of crime, progressing from simple associations to more robust and interpretable results.

5.1 Pooled OLS (All vs Census)

The Pooled OLS models were included to examine the relationship between crime rates and selected demographic, economic and social factors across LGAs. This model is useful because it provides a baseline relationship before adding fixed effects or more complex model structures. The "All" model uses the full dataset, including filled-in values across all available years, while the "Census" model only uses the census years 2011, 2016 and 2021 and has missing values for various other variables (described in detail in Data Cleaning). This allows us to check whether the filled-in data produces results that are broadly consistent with the real census-year data.

The pooled OLS model can be expressed as:

$$CrimeRate_{it} = \beta_0 + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \cdots + \beta_k X_{k,it} + \varepsilon_{it}$$

where i represents the LGA, $t$represents the year, $CrimeRate_{it}$ is the recorded offence rate per 100,000 population for LGA i in year t, $X_{k,it}$ represents the k-th explanatory variable, β_k represents its corresponding coefficient, and ε_{it}captures unexplained variation.
The dependent variable for both models is the crime rate per 100,000 population. The explanatory variables include weighted average age, median income, unemployment rate, bachelor's degree percentage, fertility rate, percentage of non-Australian citizens, male ratio, and population density. These variables were selected because they capture key demographic, economic and population-related factors that may influence crime rates. Other variables were not included where they overlapped strongly with existing variables or created multicollinearity concerns, such as using both population and density together, or using multiple education categories at the same time.

The results show that the full dataset and census-only dataset produce very similar coefficient directions and significance patterns. Bachelor's degree percentage is negative and highly significant in both models, while fertility rate is also negative and significant. The percentage of non-Australian citizens is positive and weakly significant in both models. Population density is significant in the full model but not in the census-only model. Overall, the similarity between the

two models suggests that the filled-in data is reasonably aligned with the census-year data and can be used for the broader analysis.

5.2 LGA Fixed Effects

The LGA Fixed Effects model was selected to control for unobserved, time-invariant differences across LGAs, such as geographic characteristics, long-term socio-economic conditions, or structural differences that do not change over time. This model is useful alongside the pooled OLS models because it isolates within-LGA variation over time, allowing us to better identify how changes in variables like income or unemployment affect crime within the same area. Unlike pooled OLS, this model removes bias from omitted variables that are constant within each LGA.

The LGA Fixed Effects model can be expressed as:

$$Crime_{it} = \alpha_i + \beta X_{it} + \varepsilon_{it}$$

where i represents the LGA, $t$represents the year, α_i captures unobserved time-invariant characteristics specific to each LGA, X_{it} represents the set of time-varying explanatory variables, β represents their corresponding coefficients, and ε_{it} captures unexplained variation.
The dependent variable used is offence count rather than crime rate, as fixed effects models absorb time-invariant characteristics such as population size and area. Using rates alongside LGA fixed effects can introduce distortion, while offence counts allow clearer interpretation of within-LGA changes. The explanatory variables remain the same as in the pooled models, focusing on demographic and economic factors. Variables that are constant within LGAs over time are not included separately, as they are perfectly collinear with the LGA fixed effects. The results show that, within LGAs over time, median income, unemployment rate, percentage of non-Australian citizens, and population density are all positive and statistically significant. This suggests that increases in these variables are associated with higher offence counts within the same LGA. In contrast, variables such as education, fertility rate, and male ratio are not statistically significant once LGA-specific effects are controlled for. Compared to the pooled

OLS results, some variables lose significance, indicating that earlier relationships were partly driven by differences between LGAs rather than changes within them.

5.3 Time Fixed Effects

Time Fixed Effects is the first model in the sequence that removes any shock that hit all LGAs simultaneously each year - a state-wide change in policing policy, a Victorian economic downturn, a change in crime recording methodology. After that removal, the coefficients reflect what is happening in the cross-section of LGAs within a given year, purged of those common time trends. The model can be expressed as:

$$Crime_{it} = \gamma_t + \beta X_{it} + \epsilon_{it}$$

Included predictors are all time-varying percentage or rate-based measures: Population; Population density per km²; Weighted average age; Postgraduate degree share; bachelor's degree share; Certificate share; Standardised death rate; Median income; Unemployment rate; Fertility rate; non-Australian citizen share; and Distance to CBD (km).

The following variables were excluded for the reasons outlined below:

- Detailed education categories were reduced to a few key levels to avoid multicollinearity, as many of these categories are highly correlated. Some categories were also removed because they represent incomplete or administrative classifications with limited interpretive value.
- Raw death counts were replaced with a standardised measure that adjusts for population size and age structure, allowing for better comparison across LGAs.
- Raw birth counts were replaced with a fertility measure, which standardises births relative to the population and improves comparability.
- Citizenship counts were replaced with percentage measures. Some percentage categories were excluded because they sum to 100%, which would create perfect multicollinearity if included together.

- Male and female population counts were excluded, as they are not normalised and largely capture the same demographic information already reflected in other variables.
- Area of LGAs was excluded because it does not change over time and therefore provides no variation within this model.

Several predictors change materially between Model A (raw count) and Model B (per 100,000). Population reverses sign, remaining positive and highly significant in Model A but becoming negative in Model B, confirming that it operates through scale. Larger LGAs record higher total offence counts, but lower crime rates once adjusted for population size.

Population density weighted average age, postgraduate education, fertility rate, non-Australian citizen share, and distance to the CBD remain statistically significant in both models, with consistent signs. This suggests that these variables have a stable relationship with crime beyond simple scaling effects and are more likely to represent genuine underlying predictors.

In contrast, certificate-level education, median income, and unemployment are significant in Model A but lose significance in Model B. This indicates that their apparent relationship with crime in the raw count model is largely correlated to population size rather than true differences in crime rates across LGAs.

Overall, the comparison between the two models highlights the importance of distinguishing between volume effects and rate effects. Variables that remain significant across both specifications; namely population density, weighted average age, postgraduate education, fertility rate, non-Australian citizen share, and distance to the CBD; provide stronger evidence of meaningful relationships with crime. In contrast, variables such as certificate-level education, median income, and unemployment lose significance when moving from Model A (raw count) to Model B (per 100,000), suggesting that their apparent effects are largely associated with population scale rather than underlying behavioural or socio-economic influences.

5.4 Combined Fixed Effects (LGA + Time)

The Combined Fixed Effects model controls for both LGA-specific effects and time effects. By removing these two dimensions, the remaining variation reflects true within-LGA changes over time, how changes in predictors within the same LGA are associated with changes in crime. This provides the strongest identification in the dataset without requiring an experimental or instrumental approach. The model can be expressed as:

$$Crime_{it} = \alpha_i + \gamma_t + \beta X_{it} + \epsilon_{it}$$

Included predictors capture time-varying demographic, economic, and population characteristics, including population size, population density, age structure, education composition, income, unemployment, births, and citizenship counts.

The following variable decisions differ from the Time Fixed Effects model:

- Distance to CBD: Excluded as it is time-invariant. Since it does not change over time within an LGA, it is fully absorbed by the LGA fixed effects and provides no usable variation.
- Area of LGAs: Excluded for the same reason, as geographic size does not vary over time and is captured entirely by the LGA fixed effects.
- Standardised death rate replaced with raw death counts: The model uses raw counts throughout for consistency with the raw offence count dependent variable. Standardised measures are more appropriate for cross-sectional comparisons, but less suitable for within-LGA analysis in this specification.
- Fertility rate replaced with births (raw count): Again, raw counts are used to maintain consistency with the scale of the dependent variable.
- Citizenship percentages replaced with raw counts: Percentage measures are useful for cross-sectional analysis, but raw counts are more appropriate here, as both dependent and independent variables are expressed in absolute terms.

- Detailed education sub-categories excluded: Highly correlated education variables and administrative categories are removed to avoid multicollinearity and improve interpretability.
- Crime rate, citizenship breakdowns, and gender count variables excluded: These variables are either redundant, highly correlated with included measures, or do not provide additional variation beyond what is already captured.

Several predictors change materially between the Time Fixed Effects and Combined Fixed Effects models, highlighting the importance of distinguishing between cross-sectional and within-LGA effects. Certificate-level education, deaths, median income, and unemployment lose statistical significance in the Combined Fixed Effects model, suggesting that their earlier relationships with crime were associated with stable differences across LGAs rather than changes within them over time.

In contrast, population, population density, weighted average age, and non-Australian citizen count remain statistically significant and directionally consistent across both models, indicating that these variables have robust relationships with crime even after controlling for both LGA and time effects. Fertility, measured through births, also remains strongly negative and significant, reinforcing its role as a key predictor of crime outcomes.

Notably, bachelor's degree share reverses sign, shifting from a positive association in the Time Fixed Effects model to a strong negative relationship in the Combined Fixed Effects model. This suggests that while high-education LGAs may appear to have higher crime levels in aggregate, increases in education within an LGA over time are associated with meaningful reductions in crime.

Overall, the Combined Fixed Effects model provides the strongest evidence of true within-LGA dynamics, showing that demographic structure and population composition are more important predictors of crime than purely economic indicators once unobserved heterogeneity is removed.

5.5 Factor Analysis

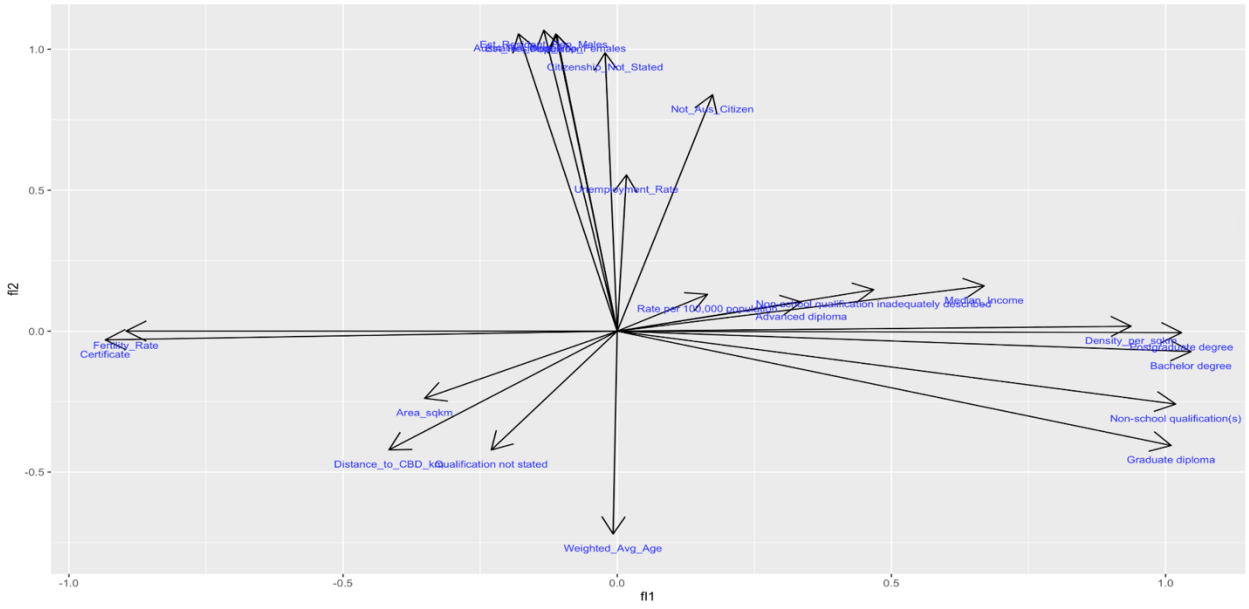
Factor analysis is employed to reduce a large set of observed variables into a smaller number of latent factors that capture the underlying correlation structure among the variables. The model can be expressed as:

$$Crime_{it} = \lambda_{j1}F_{1it} + \cdots + \lambda_{it}F + \epsilon_{it}$$

where i represents the LGA, t denotes time, F are the common factors, F_{rit} is the r[th] factor impacting observation 'it' and λ_{jr} are the factor loadings for factor r in the data dimension j, and ϵ_{it}captures the unexplained variation. Here r is the number of factors in the model and j is the number dimensions.

Three factors were extracted using Promax rotation, identifying two dominant latent dimensions, population scale and socioeconomic status, which together explain approximately 78.5% of the total variance in the dataset. This indicates that the model is effective in summarising the relationships across the explanatory variables.

Factor Analysis Results



The results of the factor analysis identify two dominant latent dimensions, population scale and socioeconomic status, which together explain 78.5% of the total variance. Despite this strong explanatory power for the overall dataset, the model does not adequately capture variation in crime.

The factor plot indicates that crime (rate per 100,000 population) is positively associated with advanced diploma, non-school qualifications (inadequately described), non-Australian citizenship, and median income, as evidenced by vectors aligned in similar directions. This suggests that higher crime rates are observed in areas characterised by greater population diversity and certain socioeconomic attributes. Conversely, crime exhibits a negative relationship with area (sq km), distance to the CBD, and qualifications not stated, implying relatively lower crime rates in larger, more remote regions with less clearly defined educational characteristics. Variables such as formal education levels, population density, and fertility rate display weak or negligible associations with crime, as indicated by their orthogonality or proximity to the origin in the factor space.

However, the high uniqueness value (69%) for the crime variable indicates that a substantial proportion of its variance remains unexplained by the extracted factors. Consequently, these relationships should be interpreted with caution, as the factor model does not provide a robust explanation of the determinants of crime.

Overall, while the model is effective in uncovering the underlying structure and interrelationships among the explanatory variables, its utility in explaining variation in crime is limited due to the weak representation of the crime variable within the factors.

5.6 LASSO Regression

The LASSO (Least Absolute Shrinkage and Selection Operator) model was included to address multicollinearity and perform variable selection. This model is useful in this context because many of the explanatory variables are highly correlated, particularly within education, population, and citizenship-related measures. Unlike pooled OLS and fixed effects models, LASSO applies a penalty to the size of coefficients, shrinking less important variables towards zero and helping identify the most relevant predictors of crime rates.

The LASSO model estimates coefficients by minimising the sum of squared errors while imposing an L_1 penalty:

$$\hat{\beta} = \arg\min_{\beta} \left[\sum_{i,t} \left(CrimeRate_{it} - \beta_0 - \sum_{j=1}^{p} \beta_j X_{j,it} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right]$$

where $CrimeRate_{it}$ represents the crime rate for LGA i in year t, $X_{j,it}$ represents predictor j, β_j represents its coefficient, $p$is the number of predictors, and λ controls the strength of the LASSO penalty. Larger values of λ impose greater coefficient shrinkage, with some coefficients potentially reduced exactly to zero.

The dependent variable used is the crime rate per 100,000 population. The explanatory variables include population density, weighted average age, bachelor's degree percentage, median income, unemployment rate, fertility rate, percentage of non-Australian citizens, and distance to the CBD. These variables were selected after removing highly correlated or overlapping variables, such as multiple education categories and population breakdowns, to ensure that LASSO could perform meaningful variable selection.

The optimal penalty parameter (λ) was selected using cross-validation. The results show that all selected variables remain in the model, indicating that each contributes some predictive value for crime rates. However, the magnitude of the coefficients highlights that fertility rate, education, and unemployment are the most influential variables. Fertility rate has a large negative effect, while bachelor's degree percentage is also negative, suggesting that higher education levels are

associated with lower crime rates. Unemployment rate is positive, indicating that increases in unemployment are associated with higher crime rates.

Other variables, including median income, population density, and distance to the CBD, have relatively small coefficients, suggesting weaker effects. The percentage of non-Australian citizens shows a positive relationship with crime, consistent with earlier models, though the magnitude is moderate.

Overall, the LASSO results are broadly consistent with the pooled OLS and fixed effects models, reinforcing that crime is influenced by a combination of socio-economic factors rather than a single dominant variable. The model also confirms the presence of multicollinearity in the data, supporting the use of regularisation methods such as LASSO and Elastic Net.

5.7 Elastic Net

Elastic Net regularisation was selected to handle the high-dimensional demographic dataset, where multicollinearity between variables such as education qualifications and citizenship measures posed a risk of overfitting and unstable coefficient estimation. The model can be expressed as:

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^{p+1}} \left[\text{SSE} + \lambda \left(\frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 + \alpha \sum_{j=1}^{p} |\beta_j| \right) \right]$$

where $\hat{\beta}$are the estimated coefficients, SSE is the prediction error, λ controls the amount of regularisation, α controls the balance between LASSO (L1) and Ridge (L2) penalties, $\beta_j$is the coefficient of predictor j, and p is the number of predictors. The model estimates coefficients by minimising the prediction error while applying both L1 and L2 penalties to reduce overfitting and improve prediction accuracy.

The model is useful alongside standard regression approaches because it simultaneously performs variable selection and coefficient shrinkage, retaining only the most informative predictors while penalising redundant or weakly associated variables. Unlike ordinary least squares, the elastic net combines both Lasso and Ridge penalties ($\alpha = 0.1$, $\lambda = 0.000416$), making it well-suited to datasets where groups of correlated variables, such as education qualification categories, are present together.

The dependent variable is crime rate per 10,000 population, measured across LGAs over multiple time periods. To address right skewness and stabilise variance, a log transformation was applied to the crime rate before modelling. As a result, all coefficients are interpreted in percentage terms; a one-unit increase in each predictor is associated with an approximate percentage change in crime rate per 10,000 population, holding all other variables constant. All predictors were standardised before model estimation, ensuring coefficients are directly comparable in magnitude and reflect each variable's relative contribution to explaining crime rate variation. Variables identified as redundant were removed before model estimation, including raw count forms of citizenship variables where percentage equivalents were retained, death-related variables, and disaggregated population by sex, to reduce duplication and improve interpretability.

The fertility rate emerges as the dominant predictor of crime rate, where a one-unit increase in fertility rate is associated with approximately a 5.3% decrease in the crime rate, suggesting that LGAs with higher fertility rates tend to have lower crime, likely reflecting the stabilising influence of family-oriented suburban communities. Educational attainment follows as the next most influential factor group, though its effects are mixed. A one-percentage-point increase in bachelor's degree holders is associated with a 2.9% decrease in crime rate, and a similar increase in non-school qualifications is associated with a 1.0% decrease, both consistent with the broader socioeconomic literature linking higher educational attainment to lower crime. Conversely, a one-percentage-point increase in inadequately described non-school qualifications, postgraduate degrees, and advanced diplomas is associated with approximately a 6.8%, 2.3%, and 1.0% increase in crime rate, respectively, which may reflect their concentration in higher-density urban areas where crime rates are also elevated, rather than a direct causal relationship. A 2.3% increase in crime rates is associated with higher proportions of individuals holding postgraduate degrees, which may reflect the presence of white-collar crimes. Among demographic variables, a one-percentage-point increase in non-Australian citizens is associated with a 2.1% decrease in crime rate, while a one-unit increase in unemployment rate is associated with a 0.9% increase; weighted average age shows a modest 0.6% decrease per additional year. By contrast, structural and size-based variables such as population, area, births, and median income are reduced to negligible effects, suggesting that once demographic composition is accounted for, the sheer scale or density of an LGA adds little additional explanatory power.

Hence, the Elastic Net model is a robust and interpretable variable selection tool for understanding the demographic drivers of area-level crime, with coefficient stability across regularisation strengths confirming the reliability of its findings.

5.8 Methodology Findings

The sequence of models shows that the relationship between crime and socio-economic factors becomes clearer as more advanced methods are applied. Pooled OLS models provide a useful baseline, with consistent signs across both the full and census datasets, supporting the validity of the interpolated data . However, their explanatory power is limited and largely reflects cross-sectional differences between LGAs.

Introducing fixed effects improves the analysis by isolating within-LGA variation. The LGA Fixed Effects model shows that variables such as unemployment, population density, and citizenship composition are positively associated with crime within LGAs, while some variables lose significance, indicating that earlier results were driven by cross-sectional differences rather than true changes over time . Time and combined fixed effects models further refine this by removing common shocks and unobserved heterogeneity, highlighting that population density, age, fertility, and citizenship measures remain consistently relevant.

Across all specifications, several variables show stable relationships with crime. Fertility rate and education (particularly bachelor's degree attainment) are consistently associated with lower crime, while unemployment and population density tend to show positive relationships. Citizenship-related variables are also repeatedly significant, though their interpretation is more complex and likely reflects broader demographic composition rather than direct effects.

Regularisation methods confirm these findings while improving model stability. LASSO retains all key variables, reinforcing their importance, while Elastic Net refines this further by reducing the influence of weaker predictors and handling multicollinearity more effectively. The Elastic Net model identifies fertility, education, unemployment, and demographic composition as the most important drivers, while reducing structural variables such as population and income to negligible effects.

Overall, the modelling sequence shows that crime is not driven by a single factor, but by a combination of demographic and socio-economic characteristics. While fixed effects models provide the strongest evidence for within-LGA relationships, Elastic Net offers the most stable and interpretable representation of the key drivers across the dataset.

6. Model Selection

Model	R^2	RMSE	MAE	MPE (%)	MAPE (%)
Pooled OLS – All Data	0.326	2890.24	2137.08	-12.75	32.53
Pooled OLS – Census Years	0.314	2952.24	2181.28	-14.08	34.13

Model	R^2	RMSE	MAE	MPE (%)	MAPE (%)
LGA Fixed Effects	0.981	912.79	577.21	-1.16	21.42
LASSO	0.357	2823.42	2086.35	-13.14	33.29
Elastic Net (Test Set)	0.777	0.21	0.17	1.23	1.94

Model selection is based primarily on MAPE (Mean Absolute Percentage Error)**,** as it provides a scale-independent measure of prediction accuracy and allows for direct comparison across models using different dependent variable specifications. RMSE and MAE are used as supporting metrics to assess the magnitude of prediction errors, while $R^2$is considered to evaluate explanatory power. ME and MPE are used only to assess bias and ensure that models are not systematically over- or under-predicting.

Based on these metrics, the Elastic Net model provides the strongest overall performance. It achieves the lowest MAPE (1.94%) and minimal bias, indicating highly accurate and stable predictions. In contrast, pooled OLS and LASSO models exhibit substantially higher MAPE values (above 30%), suggesting weaker predictive performance despite moderate explanatory power. While the LGA Fixed Effects model shows a very high $R^2$and lower error metrics, these results are not directly comparable due to the use of offence counts rather than crime rates, and because the model captures within-LGA variation rather than overall predictive accuracy.

Overall, Elastic Net is selected as the preferred model for prediction due to its superior out-of-sample accuracy and ability to handle multicollinearity, while fixed effects models remain more appropriate for interpreting within-LGA relationships.

7. Limitations

This study has several limitations related to data availability, model specification, and interpretation.

First, the dataset contains incomplete and uneven data coverage across variables. Key variables such as education attainment and citizenship composition are only available in census years, requiring interpolation to construct a balanced panel. While validation checks suggest consistency, this approach assumes that demographic characteristics change gradually over time, which may not fully capture short-term fluctuations.

Second, some variables are measured at different frequencies or levels of aggregation**.** For example, unemployment rates are assigned at a regional level rather than the LGA level, which may introduce measurement error and reduce precision in estimating local effects.

Third, there are model comparability limitations**.** Different models use different dependent variables (crime rate vs offence count) and transformations (e.g. log-transformed Elastic Net), meaning that fit statistics such as RMSE and MAE are not directly comparable across all specifications. As a result, model selection relies more heavily on scale-independent measures such as MAPE.

Fourth, while fixed effects models control for time-invariant heterogeneity, they cannot account for time-varying omitted variables**,** such as changes in policing intensity, policy interventions, or social programs, which may influence crime but are not included in the dataset.

Fifth, regularisation methods such as LASSO and Elastic Net improve prediction and handle multicollinearity, but they are less suited for causal interpretation**.** Coefficient estimates are biased due to shrinkage, and relationships should be interpreted as predictive rather than causal.

Finally, the analysis is limited to LGA-level aggregate data, which may mask important variation within LGAs and introduces the risk of ecological fallacy. Individual-level drivers of crime cannot be observed, and results should be interpreted as area-level associations rather than individual behaviour.

8. Conclusion

This paper examines the determinants of crime across Victorian LGAs using a range of econometric models applied to panel data from 2011 to 2024. The results show that crime is influenced by a combination of demographic structure and labour market conditions rather than a single dominant variable.

Across all models, several factors emerge as consistently important. Fertility rate and education are strongly associated with lower crime, suggesting that more stable, family-oriented and higher-educated communities tend to experience lower crime rates. In contrast, unemployment and population density are positively associated with crime, indicating that economic stress and urban concentration play a role in increasing crime levels. Citizenship composition also appears significant across models, though this likely reflects broader demographic and urban patterns rather than a direct causal relationship.

Among the models, Elastic Net provides the strongest overall performance, offering both high predictive accuracy and stability in the presence of multicollinearity. While fixed effects models are more appropriate for understanding within-LGA dynamics and isolating causal relationships, Elastic Net is better suited for identifying the most relevant predictors and capturing the overall structure of the data.

Overall, the findings suggest that crime in Victoria is shaped by underlying demographic and socio-economic conditions, particularly education, fertility, and labour market outcomes. These results highlight the importance of policies that support education, employment, and community stability in reducing crime at the local level.

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Appendix

Table 01: Variable Data Collection

Variable Group	Variables	Availability	Missing Periods / Notes
Geographic	Area, Distance to CBD	Full (2011–2024)	Time-invariant (no missing, constant over time)
Population	Population, Density, Age	Full (2011–2024)	No major gaps
Education (Census)	All education categories	2011, 2016, 2021	Missing all non-census years → filled using last observation
Citizenship (Census)	Citizenship counts & %	2011, 2016, 2021	Missing non-census years → filled using last observation
Mortality	Deaths, Standardised death rate	2011–2024	Death rate missing 2011–2012
Income	Median income	2016–2024	Missing 2011–2015
Labour Market	Unemployment rate	2012–2024	Missing 2011 (regional mapping applied)
Fertility	Births, Fertility rate	2018–2024	Missing 2011–2017
Gender	Male & Female population	2019–2024	Missing 2011–2018
Other Census Variables	Housing price, businesses, land, jobs	Limited years only	Excluded due to insufficient time coverage
Regional Mapping	Unemployment (regional → LGA)	Full after mapping	LGAs mapped to regions

Table 02: Pooled OLS Results for All Data and Census-Year Data

Variable	Pooled OLS: All	Pooled OLS: Census
Intercept	47051.12*	42338.02*
Weighted Average Age	-58.54	-4.65
Median Income	0.04	0.04
Unemployment Rate	-109.57	165.20
Bachelor's degree (%)	-530.64***	-475.95***
Fertility Rate	-4007.05**	-3960.37*
Non-Australian Citizen (%)	250.70.	289.51.
Male Ratio	-54308.16	-53729.35
Density per sqkm	1.24*	0.98
Observations	1,092	234
Adjusted R^2	0.321	0.289
RMSE	2890.2	2952.2

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 03: LGA Fixed Effects Model (Offence Count)

Variable	Coefficient
Weighted Average Age	-156.04
Median Income	0.10***
Unemployment Rate	252.12***
Bachelor's degree (%)	-46.16
Fertility Rate	5.82
Non-Australian Citizen (%)	284.85***
Male Ratio	9986.85
Density per sqkm	2.89***
Observations	1,092
LGA Fixed Effects	Yes
Within R^2	0.363
Adjusted R^2	0.979
RMSE	912.8

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 04: Time Fixed Effects Model A (Raw Count)

Variable	Coefficient (β)	Std. Error	Significance
Population	0.0505	0.0018	***
Population density (per km²)	0.8524	0.1766	***
Weighted average age	-313.47	47.04	***
Postgraduate degree (%)	-462.84	115.97	***
Bachelor's degree (%)	120.89	69.06	*
Certificate (%)	201.53	47.48	***
Standardised death rate	1577.80	198.79	***
Median income	-0.1227	0.0290	***
Unemployment rate (%)	201.05	117.35	*
Fertility rate	-4389.9	557.49	***
Non-Australian citizens (%)	444.21	35.40	***
Distance to CBD (km)	4.8114	1.1613	***

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 05: Time Fixed Effects Model a Fit Statistics (Raw Counts)

Within R^2	0.416
Between R^2	0.871
Overall R^2	0.849
Observations	1,092
F-statistic	162.7
F p-value	<0.001

Table 06: Time Fixed Effects Model B (per 100,000)

Variable	Coefficient (β)	Std. Error	Significance
Population	-0.0115	0.0018	***
Population density (per km²)	0.9312	0.1721	***
Weighted average age	-380.09	45.83	***
Postgraduate degree (%)	-494.73	112.99	***
Bachelor's degree (%)	-42.86	67.29	—
Certificate (%)	17.170	46.26	—
Standardised death rate	1649.90	193.68	***
Median income	-0.0343	0.0283	—
Unemployment rate (%)	61.758	114.33	—
Fertility rate	-5678.2	543.16	***
Non-Australian citizens (%)	251.12	34.49	***
Distance to CBD (km)	6.5182	1.1315	***

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 07: Time Fixed Effects Model B Fit Statistics (per 100,000)

Within R^2	0.388
Between R^2	0.459
Overall R^2	0.440
Observations	1,092
F-statistic	145.9
F p-value	<0.001

Table 08: Combined Fixed Effects Model

Variable	Coefficient (β)	Std. Error	Significance
Population	0.0655	0.0050	***
Population density (per km²)	0.9213	0.3162	***
Weighted average age	-250.63	71.21	***
Postgraduate degree (%)	113.96	62.31	*
Bachelor's degree (%)	-245.52	64.19	***
Certificate (%)	34.010	59.96	—
Deaths (raw count)	0.5616	0.5581	—
Median income	-0.0034	0.0158	—
Unemployment rate (%)	67.181	75.77	—
Births (raw count)	-1.3141	0.3432	***
Non-Australian citizens (count)	-0.0545	0.0148	***

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.10$.

Table 09: Combined Fixed Effects Model Fit Statistics

Within R^2	0.614
Between R^2	0.001
Overall R^2	0.007
Observations	1,092
F-statistic	59.64
F p-value	<0.001

Table 10: LASSO Regression Results ($\lambda = 2.31$)

Variable	Coefficient
Density per sqkm	0.92
Weighted Average Age	-139.05
Bachelor's degree (%)	-363.11
Median Income	0.045
Unemployment Rate	149.76
Fertility Rate	-4984.59
Non-Australian Citizen (%)	213.23
Distance to CBD (km)	9.60

Table 11: Elastic Net Results with their associations to the crime rate

Variables	Coefficients	Association with Crime Rate
Non-school qualification inadequately described	0.06816045	Positive
Postgraduate degree	0.02293373	Positive
Year	0.01649303	Positive
Advanced diploma	0.00963204	Positive
Unemployment Rate	0.00878164	Positive
Qualification not stated	0.00714035	Positive
Fertility Rate	-0.05434732	Negative
Bachelor's degree	-0.02934635	Negative
Non-Australian Citizen Percentage	-0.02084119	Negative
Non-school qualification(s)	-0.009567171	Negative
Weighted Average Age	-0.006112146	Negative
Density per sqkm	0.000112696	Negligible effect
Births	8.19562E-06	Negligible effect
Area sqkm	0.000003253	Negligible effect
Median Income	2.12765E-06	Negligible effect
Population	5.49263E-07	Negligible effect

Certificate	-0.00560921	Negligible effect
Graduate diploma	-0.004444804	Negligible effect
Distance to CBD km	-0.000226821	Negligible effect

Table 12: Elastic Net Model Fit Results

Metric	Value
R^2	0.7770
ME	0.1135
RMSE	0.2096
MAE	0.1723
MPE (%)	1.2272
MAPE (%)	1.9418